Fleet Efficiency & Trip Duration Prediction (EDA + Machine Learning)

Overview

This project focuses on analyzing large-scale real-world fleet data to uncover operational inefficiencies and build a machine learning model to predict trip duration. The goal is to simulate how modern fleet platforms optimize operations using data-driven insights.

Business Problem

Fleet operators often face challenges such as:

- Unpredictable trip durations
- High vehicle idle time
- Inefficient route planning
- Poor demand forecasting

These issues lead to:

- Increased operational costs
- Reduced fleet utilization
- Lower customer satisfaction

Solution

Developed an end-to-end data pipeline to:

- Analyze trip patterns using Exploratory Data Analysis (EDA)
- Clean and preprocess real-world noisy data
- Engineer meaningful features from raw telemetry
- Build a machine learning model to predict trip duration

Dataset

- Source: NYC Taxi Trip Data (official Parquet dataset)
    - Link: "https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page"
- Type: Real-world, large-scale transportation data
- Features include:

- Pickup & drop-off timestamps
- Trip distance
- Passenger count
- Fare amount
- Location IDs

Data Preprocessing

Performed extensive data cleaning to handle real-world challenges:

- Removed trips with zero or invalid distance
- Filtered unrealistic trip durations (<1 min or >120 min)
- Handled missing values
- Converted datetime fields into usable formats

Exploratory Data Analysis (EDA)

EDA was conducted to uncover operational insights:

Key Findings

- 📈 **Peak Demand Hours**: High trip volume observed during morning (8–10 AM) and evening (5–7 PM)
- 🚚 **Fleet Utilization Patterns**: Lower utilization during mid-day hours indicating idle time
- 📏 **Distance vs Duration**: Strong relationship between trip distance and duration
- ⚠️ **Data Anomalies**: Identified outliers such as extremely long or short trips

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Feature Engineering

Created meaningful features to improve model performance:

- Extracted **hour, day of week, and weekend indicators**
- Created **trip speed feature (distance / duration)**
- Converted datetime into actionable temporal features

Model Development

Model Used:

- Random Forest Regressor

Why Random Forest?

- Handles non-linear relationships effectively
- Robust to noise and outliers
- Performs well on structured/tabular data

Model Evaluation

- **Mean Absolute Error (MAE):** ~4–5 minutes
- **R^2 Score:** Indicates strong predictive performance

Interpretation:

The model predicts trip duration with an average error of ~4–5 minutes, which is acceptable for real-time ETA estimation in fleet systems.

Feature Importance

Key drivers of trip duration:

- Trip distance (most important)
- Time of day (traffic patterns)
- Day of week (weekday vs weekend trends)

Business Impact

This solution can help fleet operators:

- 🚀 Improve ETA accuracy
- 📉 Reduce idle time
- 📍 Optimize driver allocation
- 💰 Increase operational efficiency

Future Improvements

- Integrate real-time streaming data

- Apply deep learning models for better accuracy
- Incorporate GPS route optimization
- Build interactive dashboard for business users

Tech Stack

- Python (Pandas, NumPy, Scikit-learn)
- Data Visualization (Seaborn, Matplotlib)
- Machine Learning (Random Forest)

Conclusion

This project demonstrates how real-world data can be leveraged to improve operational efficiency using data science and machine learning. It highlights the importance of combining EDA, feature engineering, and predictive modeling to deliver business value.

Author

Sankarakumar Jeevarathinam
AI & Data Product Enthusiast | Machine Learning Practitioner